

Ferroptosis Involvement in Glioblastoma Treatment.

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Glioblastoma multiforme (GBM) is one of the deadliest brain tumors. Current standard therapy includes tumor resection surgery followed by radiotherapy and chemotherapy. Due to the tumors invasive nature, recurrences are almost a certainty, giving the patients after diagnosis only a 12-15 months average survival time. Therefore, there is a dire need of finding new therapies that could potentially improve patient outcomes. Ferroptosis is a newly described form of cell death with several implications in cancer, among which GBM. Agents that target different molecules involved in ferroptosis and that stimulate this process have been described as potentially adjuvant anti-cancer treatment options. In GBM, ferroptosis stimulation inhibits tumor growth, improves patient survival, and increases the efficacy of radiation and chemotherapy. This review provides an overview of the current knowledge regarding ferroptosis modulation in GBM.

Glioblastoma multiforme of spinal cord - Case series in a tertiary cancer centre.

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Primary spinal cord glioblastoma multiforme (GBM) is a rare clinical condition and is often associated with a dismal prognosis. The standard treatment is maximal safe surgery followed by adjuvant radiotherapy and chemotherapy. Despite such aggressive treatments, the median survival is estimated to be around 15 months in several studies. We report three patients with primary spinal GBM who received treatment in our institute from 2012 to 2019. Among the three, one patient is on long-term follow-up with no evidence of disease, another patient succumbed to the illness and the third patient is having stable disease. Although primary spinal GBM is usually associated with a dismal prognosis, our case series shows a subset of patients will have a favorable outcome with the protocol treatment.

Prodigiosin stimulates endoplasmic reticulum stress and induces autophagic cell death in glioblastoma cells.

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Prodigiosin, a secondary metabolite isolated from marine *Vibrio* sp., has antimicrobial and anticancer properties. This study investigated the cell death mechanism of prodigiosin in glioblastoma. Glioblastoma multiforme (GBM) is an aggressive primary cancer of the central nervous system. Despite treatment, or standard therapy, the median survival of glioblastoma patients is about 14.6 month. The results of the present study clearly showed that prodigiosin significantly reduced the cell viability and neurosphere formation ability of U87MG and GBM8401 human glioblastoma cell lines. Moreover, prodigiosin with fluorescence signals was detected in the endoplasmic reticulum and found to induce excessive levels of autophagy. These findings were confirmed by observation of LC3 puncta formation and acridine orange staining. Furthermore, prodigiosin caused cell death by activating the JNK pathway and decreasing the AKT/mTOR pathway in glioblastoma cells. Moreover, we found that the autophagy inhibitor 3-methyladenine reversed prodigiosin induced autophagic cell death. These findings of this study suggest that prodigiosin induces autophagic cell death and apoptosis in glioblastoma cells.

Temozolomide in Combination With NF- κ B Inhibitor Significantly Disrupts the Glioblastoma Multiforme Spheroid Formation.

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Glioblastoma multiforme (GBM) is the most common malignant primary brain tumor, accounting for 50% of all cases. GBM patients have a five-year survival rate of merely 5.6% and a median overall survival of 14.6 months with the "Stupp" regimen, 20.9 months with tumor treatment fields (TTF, OptuneR) in patients who participated in clinical trials, and 11 months for all GBM patients prior to TTF use. Objective: Our group recently developed a brain cancer chip which generates tumor spheroids, and provides large-scale assessments on the response of tumor cells to various concentrations and combinations of drugs. This platform could optimize the use of tumor samples derived from GBM patients to provide valuable insight on the tumor growth and responses to drug therapies. To minimize any sample loss in vitro, we improved our brain cancer chip system by adding an additional laminar flow distribution layer, which reduces sample loss during cell seeding and prevents spheroids from escaping from the microwells. Methods: In this study, we cultured 3D spheroids from GBM cell lines and patient-derived GBM cells in vitro, and investigated the effect of the combination of Temozolomide and nuclear factor- κ B inhibitor on tumor growth. Results: Our study revealed that these drugs have synergistic effects in inhibiting spheroid formation when used in combination. Conclusions: These results suggest that the brain cancer chip enables large-scale, inexpensive and sample-effective drug screening to 3D cancer tumors in vitro, and could be applied to related tissue engineering drug screening studies.

Role of Chimeric Antigen Receptor T Cell Therapy in Glioblastoma Multiforme.

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Glioblastoma multiforme (GBM) is the most common primary malignant cancer of brain, which is extremely aggressive and carries a dreadful prognosis. Current treatment protocol runs around radiotherapy, surgical resection, and temozolomide with median overall survival of around 12-15 months. Due to its heterogeneity and mutational load, immunotherapy with chimeric antigen receptor (CAR) T cell therapy can be a promising treatment option for recurrent glioblastoma. Initial phase 1 studies have shown that this therapy is safe without dose-limiting side effects and it also has a better clinical outcome. Therefore, CAR T cell therapy can be a great future tool in our armamentarium to treat advanced GBM. In this article, we have explained the structure, mechanism of action, and rationale of CAR T cell therapy in GBM; we also discussed various antigenic targets and clinical outcome of initial studies of this novel therapy.

Cannabidiol May Prolong Survival in Patients With Glioblastoma Multiforme.

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Glioblastoma multiforme (GBM) is a relatively rare type of brain tumour with an incidence rate around 6 per 100,000. Even with the widely practiced combination of radiotherapy with adjuvant temozolomide, the median overall survival remains low with just 13.5 to 16 months after diagnosis. We retrospectively

reviewed the survival of a cohort of 15 consecutive, unselected patients with histopathologically confirmed glioblastoma multiforme (GBM) who received CBD (400 to 600 mg orally per day) in addition to standard therapy (maximum resection of the tumour followed by radio-chemotherapy). Of 15 patients, seven (46.7%) are now living for at least 24 months, and four (26.7%) for at least 36 months. This is more than twice as long as has been previously reported in the literature. The mean overall survival is currently 24.2 months (median 21 months). CBD is a well supported co-medication and seems to prolong the survival of patients with glioblastoma multiforme.

Exploring the Functional Roles of Telomere Maintenance 2 in the Tumorigenesis of Glioblastoma Multiforme and Drug Responsiveness to Temozolomide.

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Glioblastoma multiforme (GBM) is a grade IV human glioma. It is the most malignant primary central nervous system tumor in adults, accounting for around 15% of intracranial neoplasms and 40-50% of all primary malignant brain tumors. However, the median survival time of GBM patients is still less than 15 months, even after treatment with surgical resection, concurrent chemoradiotherapy, and adjuvant chemotherapy with temozolomide (TMZ). Telomere maintenance 2 (TELO2) mRNA is highly expressed in high-grade glioma patients, and its expression correlates with shorter survival outcomes. Hence, it is urgent to address the functional role of TELO2 in the tumorigenesis and TMZ treatment of GBM. In this study, we knocked down TELO2 mRNA in GBM8401 cells, a grade IV GBM, compared with TELO2 mRNA overexpression in human embryonic glial SVG p12 cells and normal human astrocyte (NHA) cells. We first analyzed the effect of TELO2 on the Elsevier pathway and Hallmark gene sets in GBM8401, SVG p12, and NHA via an mRNA array analysis. Later, we further examined and analyzed the relationship between TELO2 and fibroblast growth factor receptor 3, cell cycle progression, epithelial-mesenchymal transition (EMT), reactive oxygen species (ROS), apoptosis, and telomerase activity. Our data showed that TELO2 is involved in several functions of GBM cells, including cell cycle progression, EMT, ROS, apoptosis, and telomerase activity. Finally, we examined the crosstalk between TELO2 and the responsiveness of TMZ or curcumin mediated through the TELO2-TTI1-TTI2 complex, the p53-dependent complex, the mitochondrial-related complex, and signaling pathways in GBM8401 cells. In summary, our work provides new insight that TELO2 might modulate target proteins mediated through the complex of phosphatidylinositol 3-kinase-related kinases in its involvement in cell cycle progression, EMT, and drug response in GBM patients.

Survival and quality of life analysis in glioblastoma multiforme with adjuvant chemoradiotherapy: a retrospective study.

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Glioblastoma multiforme (GBM) is the most common and aggressive primary intracranial tumor. Despite modern therapies, it is still fatal with tremendously poor prognosis with a median survival of 14 months. Even though mean survival and progression-free survival (PFS) are considered as primary response measure, it is important to assess the effects of therapies on disease burden and health-related quality of life (HRQoL). Changes in quality of life (QoL) indicates the impact of cytotoxic therapy and may aid in defining response in the absence of quantifiable endpoints like tumor regression. The objective was to

assess 2-year survival and quality of life in GBM patients who underwent primary surgery followed by chemo-radiotherapy and 6-month adjuvant chemotherapy with temozolomide. Single-institution retrospective study of 60 patients of GBM from 2015 to 2017. Data regarding patient factors, disease factors, and treatment factors were collected and survival was calculated. 60 patients with GBM were analyzed, male to female ratio was 1.6:1. Patients most commonly presented with headache. Most common tumour site is the frontal lobe. The median overall survival (OS) was 10 months. The 1-year and 2-year survival rates were 30% and 6.7%, respectively. Compared to before surgery patients have showed improved emotional, social and role functioning in Post radiotherapy period. There was a decrease in symptoms like pain, headache and seizures. OS and QoL in GBM patients remains poor despite constant research and studies. Maximum safe resection followed by adjuvant temozolomide has shown improvement in OS. Post-surgery and adjuvant radiotherapy patients have shown a decrease in symptoms and better QoL.

Improved median survival for glioblastoma multiforme following introduction of adjuvant temozolomide chemotherapy.

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The use of adjuvant temozolomide (TMZ) in patients managed with surgery and adjuvant radiation therapy (RT) for glioblastoma multiforme (GBM) has been demonstrated to improve median and 2-year survival in a recent large international multicentre study. To confirm this result in routine clinical practice, an audit of the management and outcome of patients with GBM at The Cancer Institute Radiation Oncology was performed. All patients with GBM managed radically at The Cancer Institute Radiation Oncology from May 2002 to 2006 were entered into a prospective database. Patient, tumour and treatment factors were analysed for association with the outcome of median survival (MS). Survival was calculated using the Kaplan-Meier technique and correlation was assessed using Cox proportional hazards regression. Forty-one patients with GBM were managed with radical intent over the 4- year period. The median age was 54 years and 66% were Eastern Cooperative Oncology Group (ECOG) 0-1 performance status. Macroscopic, subtotal and biopsy alone procedures were performed in 61%, 29% and 10% of patients, respectively. The median time from surgery to RT was 26 days. Adjuvant TMZ was used in 44% of patients (n = 18). The MS of the total group was 13.6 months, with a 24% 2-year overall survival. The use of TMZ was associated with improved MS (19.6 versus 12.8 months; $P = 0.035$) and improved 2-year survival (43% versus 0%). A requirement of dexamethasone dose greater than 4 mg at the end of RT ($P = 0.012$) was associated with worse survival, but there was no association of MS with age, ECOG, tumour size or extent of surgery. The median and 2-year survival outcomes are comparable to the results of the European Multicentre Study and justify the continued use of TMZ in routine clinical practice.

One-carbon-mediated purine synthesis underlies temozolomide resistance in glioblastoma.

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Glioblastoma accounts for nearly half of all primary malignant brain tumors in adults, and despite an aggressive standard of care, including excisional surgery and adjuvant chemoradiation, recurrence

remains universal, with an overall median survival of 14.6 months. Recent work has revealed the importance of passenger mutations as critical mediators of metabolic adaptation in cancer progression. In our previous work, we identified a role for the epigenetic modifier ID-1 in temozolomide resistance in glioblastoma. Here, we show that ID-1-mediated glioblastoma tumourigenesis is accompanied by upregulation of one-carbon (1-C) mediated de novo purine synthesis. ID-1 knockout results in a significant reduction in the expression of 1-C metabolism and purine synthesis enzymes. Analysis of glioblastoma surgical specimens at initial presentation and recurrence reveals that 1-C purine synthesis metabolic enzymes are enriched in recurrent glioblastoma and that their expression correlates with a shorter time to tumor recurrence. Further, we show that the 1-C metabolic phenotype underlies proliferative capacity and temozolomide resistance in glioblastoma cells. Supplementation with exogenous purines restores proliferation in ID-1-deficient cells, while inhibition of purine synthesis with AICAR sensitizes temozolomide-resistant glioblastoma cells to temozolomide chemotherapy. Our data suggest that the metabolic phenotype observed in treatment-resistant glioma cells is a potential therapeutic target in glioblastoma.
